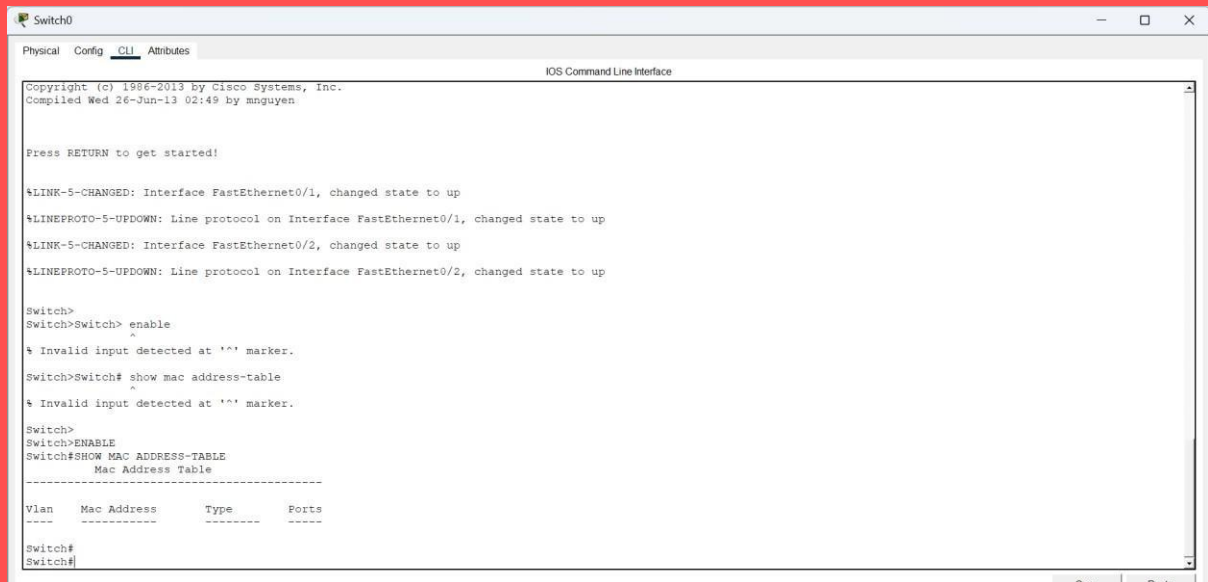


SWITCHING BASICS THEORY

1.Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty. ○



```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
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Compiled Wed 26-Jun-13 02:49 by mnquyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>
Switch>Switch> enable
Switch>Switch# show mac address-table
Switch>Switch#
Switch>
Switch>ENABLE
Switch#SHOW MAC ADDRESS-TABLE
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
-----
Switch#
Switch#
```

Initially, the switch MAC address table is empty because the switch has not yet learned the MAC addresses of the connected PCs. The MAC addresses will be learned automatically when the PCs start sending network traffic.

**2.Send a ping from PC1 to PC2 in your simulated network, then immediately run 'show mac address-table' on the switch and note how the MAC addresses of both PCs appear in the table.

Hint: Try pinging in both directions to see how the table updates.**

○

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

$LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
$LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
$LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>
Switch>Switch> enable
^
% Invalid input detected at '^' marker.
Switch>Switch# show mac address-table
^
% Invalid input detected at '^' marker.

Switch>
Switch>ENABLE
Switch#SHOW MAC ADDRESS-TABLE
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
Switch#
Switch#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
1       0060.5cca.b376    DYNAMIC   Fa0/1
1       00d0.ba00.ab13    DYNAMIC   Fa0/2
Switch#
```

After sending a ping between the two PCs, the switch learned their source MAC addresses and added them dynamically to the MAC address table. The MAC addresses are associated with FastEthernet0/1 and FastEthernet0/2 respectively.

3.Add a third PC to your network, connect it to the switch, and ping from PC3 to PC1. Observe and record how the MAC address table changes after this communication.



```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
1       0060.5cca.b376    DYNAMIC   Fa0/1
1       00d0.ba00.ab13    DYNAMIC   Fa0/2
Switch#SHOW MAC ADDRESS-TABLE
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
1       0060.5cca.b376    DYNAMIC   Fa0/1
1       00d0.ba00.ab13    DYNAMIC   Fa0/2
Switch#SHOW MAC ADDRESS-TABLE
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
1       0060.5cca.b376    DYNAMIC   Fa0/1
1       00d0.587c.21ba    DYNAMIC   Fa0/3
1       00d0.ba00.ab13    DYNAMIC   Fa0/2
Switch#
```

After adding the third PC and pinging PC1, the switch learned the third PC's MAC address dynamically. The MAC address table was updated with a new entry on FastEthernet0/3, while the existing MAC addresses remained associated with Fa0/1 and Fa0/2. This demonstrates how a switch dynamically learns MAC addresses from network traffic.

4. Suppose you see the following MAC address table on a switch: Port 1 - 00:11:22:33:44:55, Port 2 - 66:77:88:99:AA:BB. If a frame arrives at the switch on Port 1 with destination MAC 66:77:88:99:AA:BB, explain step by step how the switch will forward this frame.

○

Suppose the MAC address table is:

Port	MAC Address
Port 1	00:11:22:33:44:55
Port 2	66:77:88:99:AA:BB

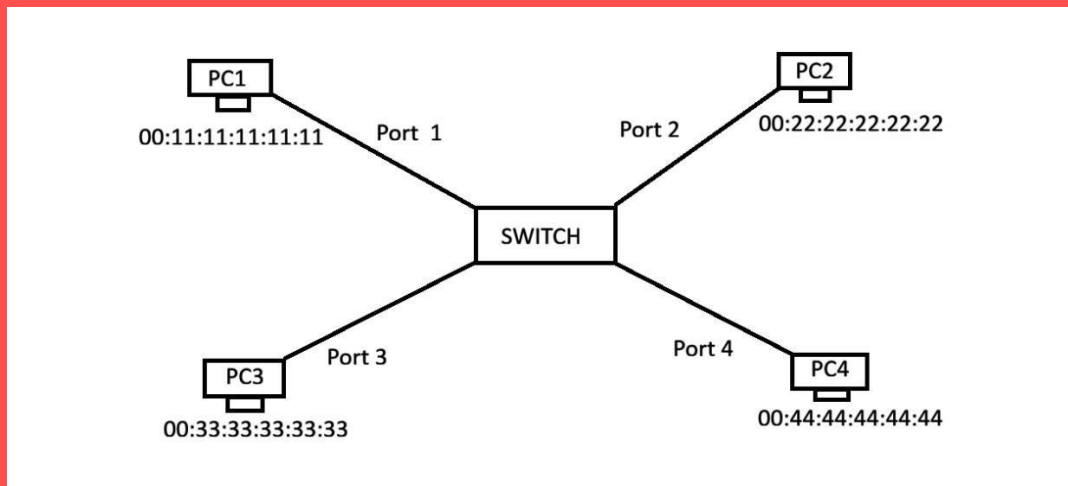
If a frame arrives on Port 1 with destination MAC 66:77:88:99:AA:BB:

1. The switch receives the frame on Port 1.
2. It checks the destination MAC address of the frame.
3. The switch searches its MAC address table and finds 66:77:88:99:AA:BB associated with Port 2.
4. Because the destination MAC is known, the switch forwards the frame only through Port 2.
5. The frame is not sent to other ports because the switch already knows the correct destination port.

**5. Draw a diagram showing how a switch learns and uses MAC addresses to forward frames in a network with 4 devices, and write a short explanation (3-4 sentences) describing the process in your own words.

Constraint: Do not copy definitions from the textbook; use your own understanding and examples.**

○



When a device sends a frame, the switch learns its MAC address and the port from which it came. If the destination MAC address is known, the switch forwards the frame only to the correct port. If the destination is unknown, the switch sends the frame to other ports. After learning the MAC addresses, the switch can forward frames directly to the correct device.